

Let X be any real valued random variable such that $a \leq X \leq b$ almost surely. Then for any $\lambda \in \mathbb{R}$,

$$\mathbb{E} \left[e^{\lambda(X - \mathbb{E}[X])} \right] \leq e^{\left(\frac{\lambda^2 (b-a)^2}{8} \right)}$$

Proof using convexity of $e^{\lambda x}$ and then AMGM inequality.

0.0.1 See also

0.0.2 References